Statistical Inference 2

Data Science in Practice

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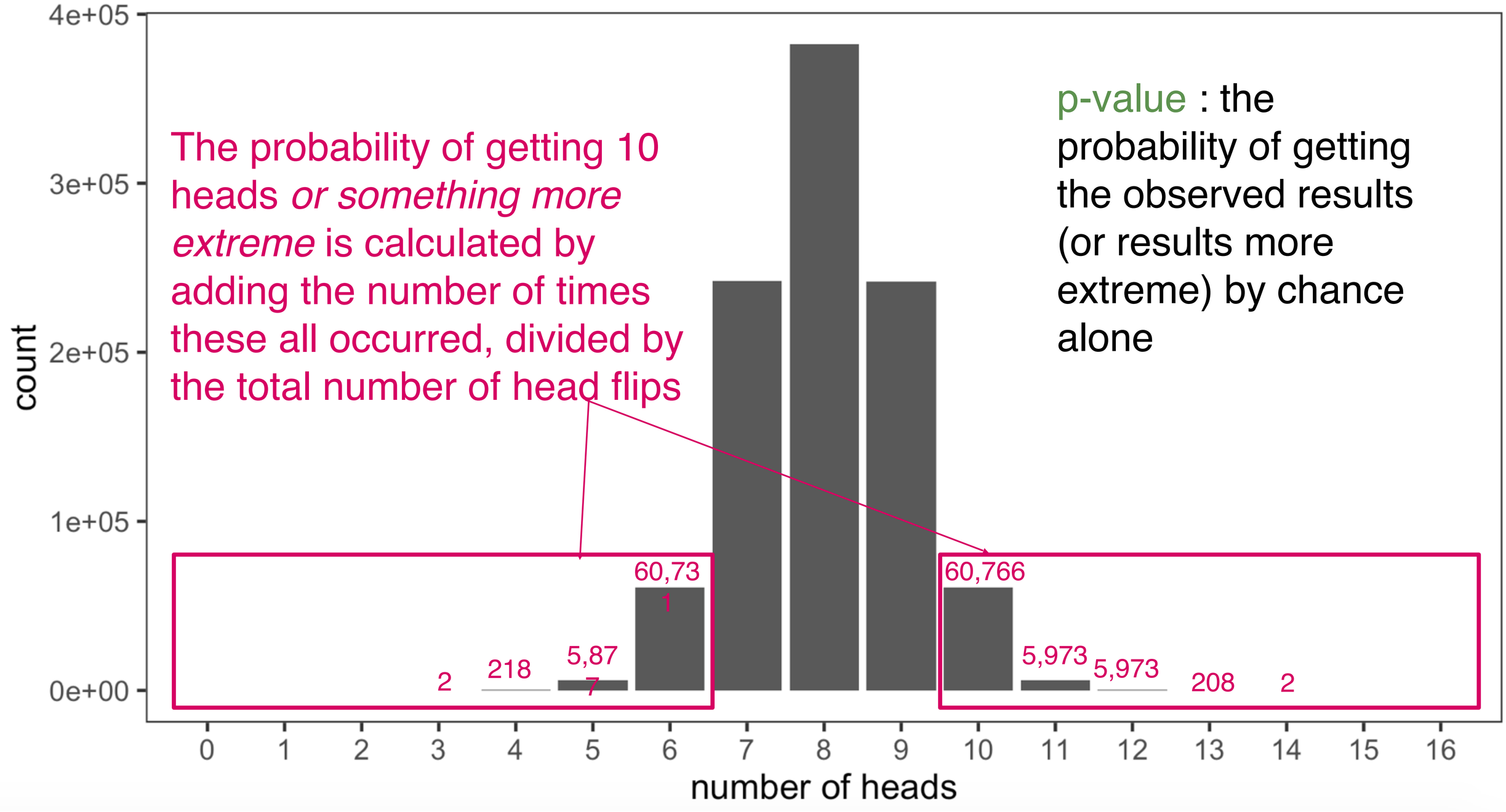


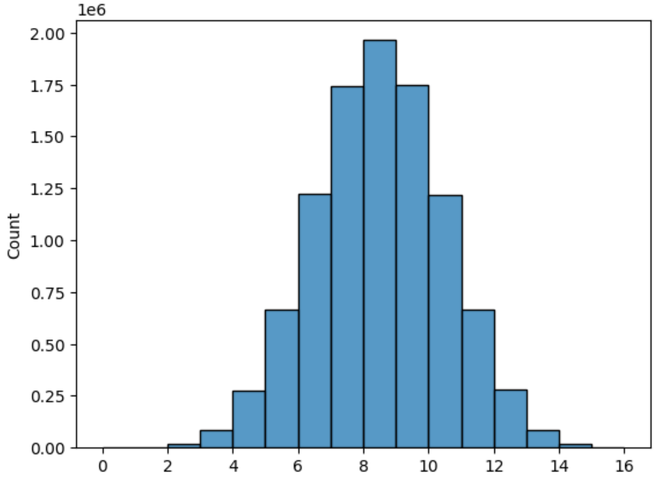
The probability of getting 10 heads *or something more extreme* is calculated by adding the number of times these all occurred, divided by the total number of head flips

p-value : the probability of getting the observed results (or results more extreme) by chance alone

Monte Carlo simulation

Making an empirical null distribution instead of a t-test

1. First, we postulate a null hypothesis (H_o), e.g. the coin we are looking at is a fair coin
2. Next, we simulate many draws from of our postulated process and compute a statistic on each one
3. Finally, we calculate the p-value of our real observed data in comparison to the simulated null distribution



P-value of 12 heads or weirder?

https://github.com/COGS108/Tutorials/blob/master/Monte_Carlo_simulation_of_coin_flip.ipynb

The interpretation of p-values in NHST

- Null hypothesis statistical testing (NHST) means we assume there is no difference
  - Sample A and sample B are compared… one of them has a statistic that is bigger than the other
- P-value represents
  - What's the likelihood of getting values *at least* this weird **given an assumption that sample A and sample B come from the same distribution**?
  - We call this the null hypothesis assumption or H_0
  - A p-value is $P(D \mid H_0)$

The interpretation of p-values in NHST

- Historically because some old dead guy said so, we take a value of p=0.05 as a threshold of “significance”
- This represents a 1/20 chance of this data under the null assumption
- What happens if you ask 20 questions using NHST?
  - Take a look at this: https://xkcd.com/882/

The interpretation of p-values in NHST

- When your data are very inconsistent with the null hypothesis (e.g. p=0.47), P values can’t determine which of the following possibilities is true:
  - The null hypothesis is false, but your sample is unusual due to random sampling.
  - The null hypothesis is false, but you don’t have enough samples to see the small difference between sample A and sample B
  - The null hypothesis is true

- When your data show the null has a high probability (e.g. p=0.001), then we can’t determine which of the following possibilities is true:
  - The null hypothesis is true, but your sample is unusual due to random sampling.
  - The null hypothesis is false (yay!)

The interpretation of p-values in NHST

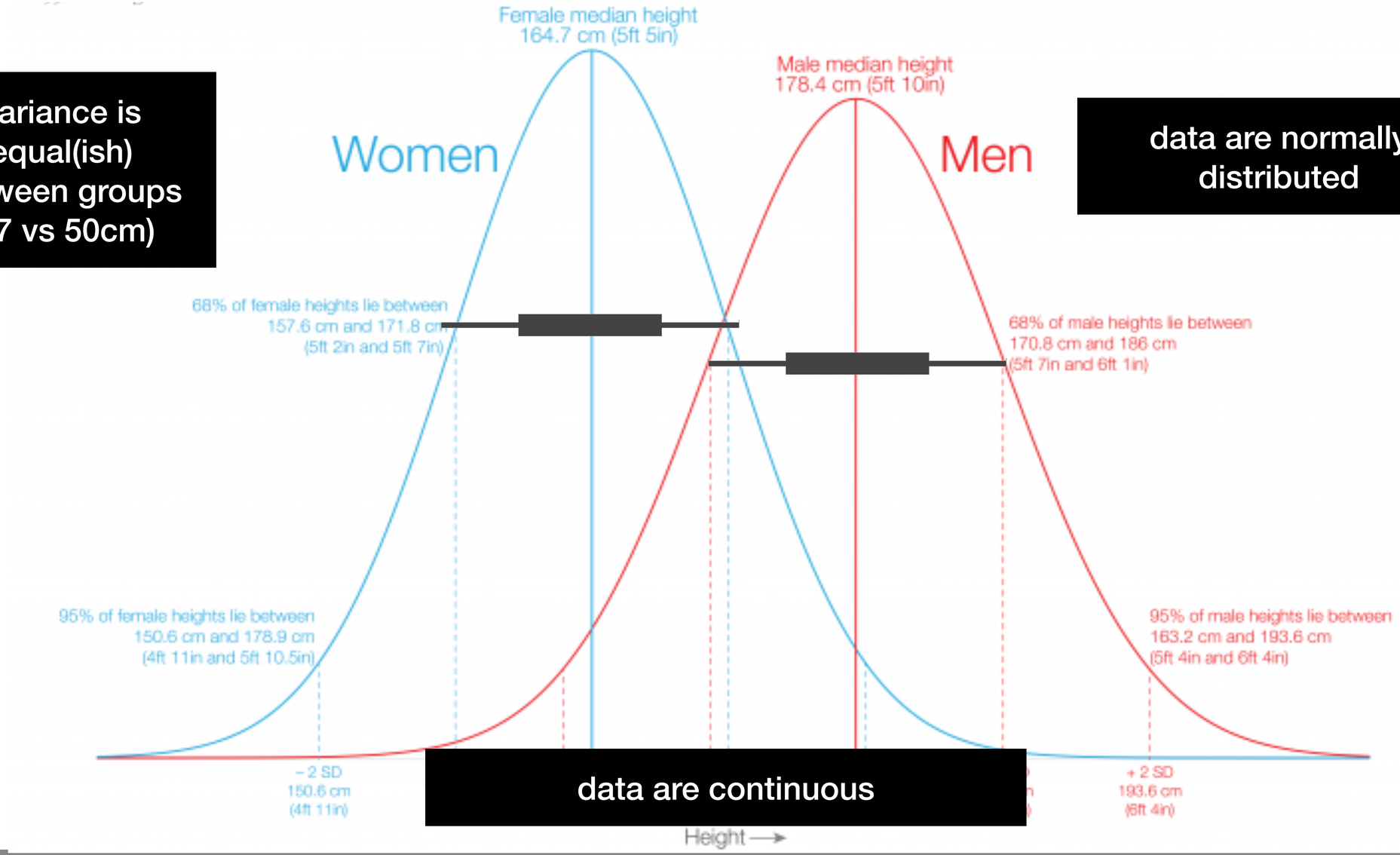
- P-value is NOT the chance of a Type I error! This is a common mistake!
  - Type 1 error: rejecting the null, when it is really true
    - AKA a false positive result
    - Notated as $P(H_0 | D)$
  - Bayesian viewpoint: What's the prior probability of the null?
    - If this is the first study ever on a topic then we know less about the truth than if this is the 36th study on a topic
    - $P(H_0 | D) = P(D | H_0) P(H_0)$
  - Empirically a p=0.05 is usually a **MUCH BIGGER Type 1** error rate
    - Can't know in reality, but simulation / math studies tell us that we can expect something like for p=0.05 a Type 1 error rate of somewhere like 23 to 50%

https://forms.gle/6MCyp7qFsaHgGKi5A

Do the heights between males and females differ?

variance is equal(ish) between groups (57 vs 50cm)

data are normally distributed

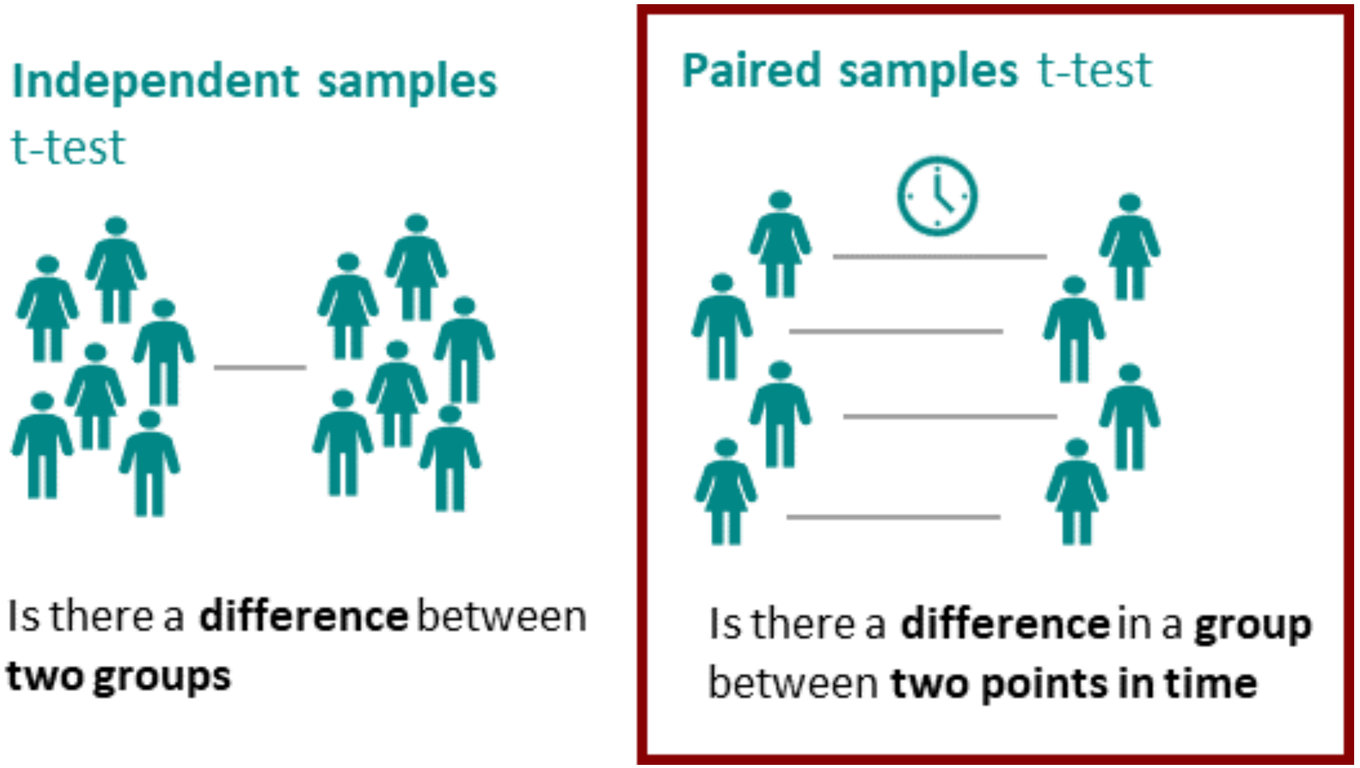


data are continuous

sample size affects statistic

N=10,000

Paired data

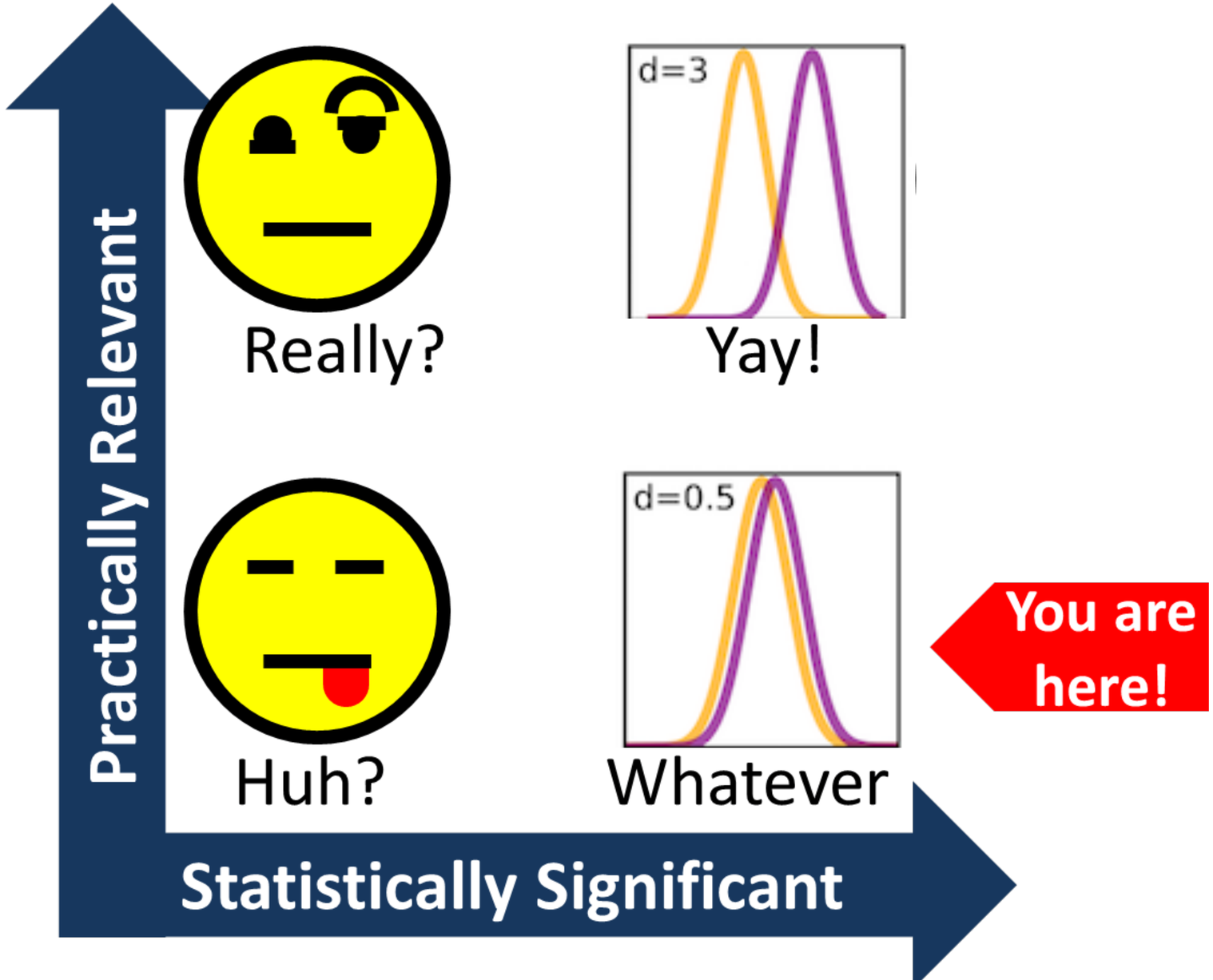


https://datatab.net/tutorial/paired-t-test

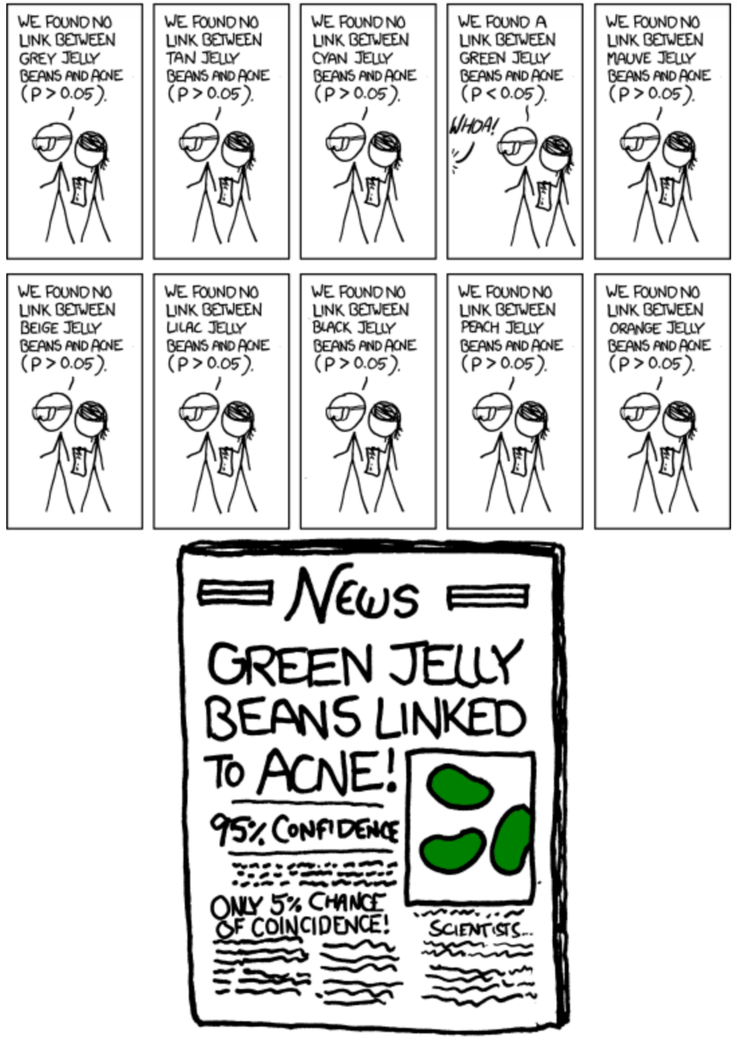
Cohen's *d*

Cohen's *d* is defined as the difference between two means divided by the pooled standard deviation

Effect sizes!



Beyond Pairwise testing: lots of t-tests



Bonferroni correction

14 languages

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From Wikipedia, the free encyclopedia

Bonferroni correction is a method to counteract the multiple comparisons problem in statistics. It is named after the mathematician Carlo Emilio Bonferroni .

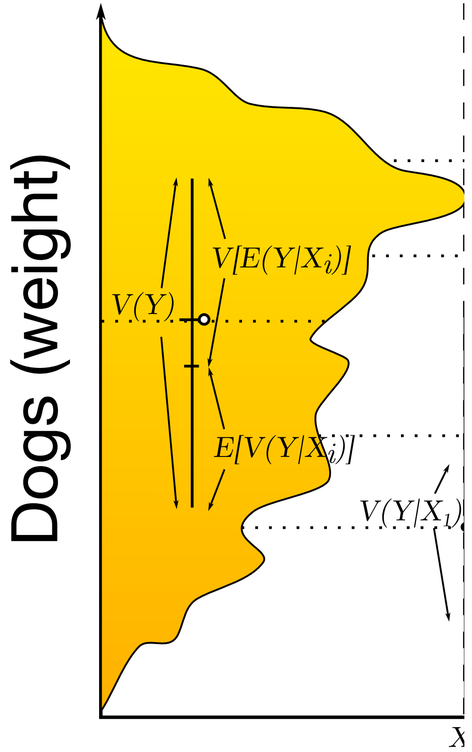
Background [edit]

Statistical hypothesis testing is based on rejecting the null hypothesis when the likelihood of the observed data would be low if the null hypothesis were true. If multiple hypotheses are tested, the probability of observing a rare event increases, and therefore, the likelihood of incorrectly rejecting a null hypothesis (i.e., making a Type I error) increases.[1]

The Bonferroni correction compensates for that increase by testing each individual hypothesis at a significance level of α/m, where α is the desired overall alpha level and m is the number of hypotheses.[2] For example, if a trial is testing $m = 20$ hypotheses with a desired overall $\alpha = 0.05$, then the Bonferroni correction would test each individual hypothesis at $\alpha = 0.05/20 = 0.0025$. The method is named for its use of the Bonferroni inequalities.[3]

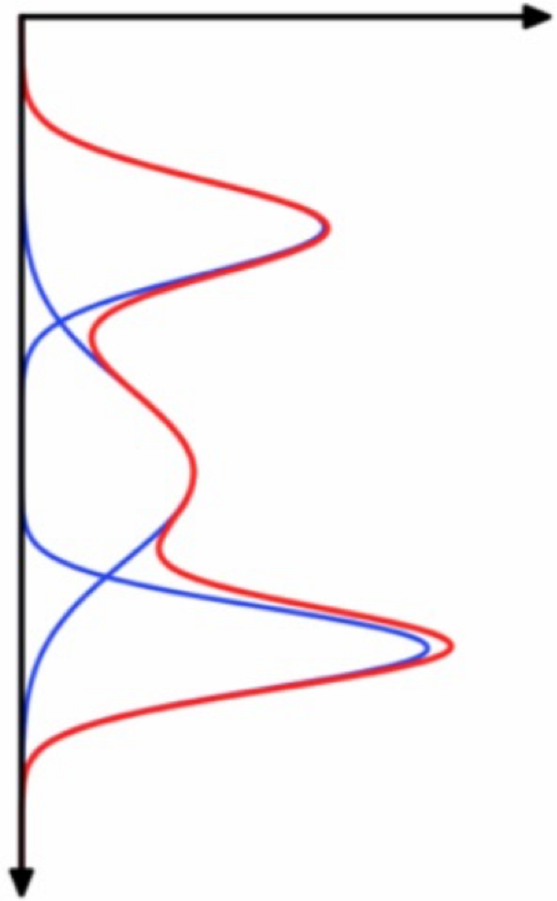
Beyond Pairwise testing: ANOVA

Same assumptions as t-test



Pretty clearly not a single normal distribution

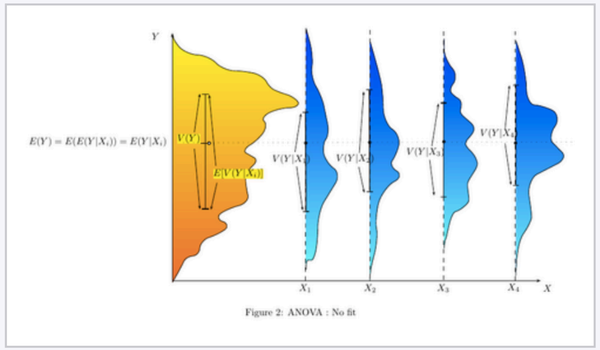
But what if its a bunch of normals slammed together?



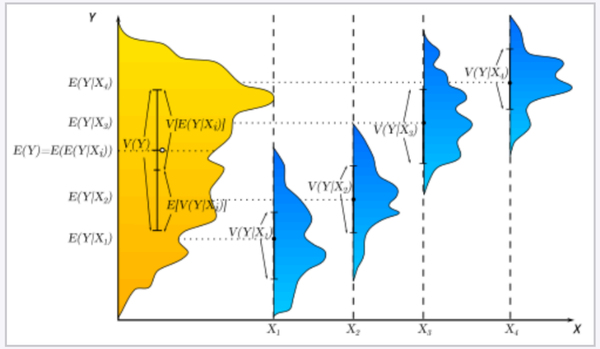
https://en.wikipedia.org/wiki/Analysis_of_variance

Beyond Pairwise testing: ANOVA

Same assumptions as t-test



No fit: Young vs old, and short-haired vs long-haired



Fair fit: Pet vs Working breed and less athletic vs more athletic

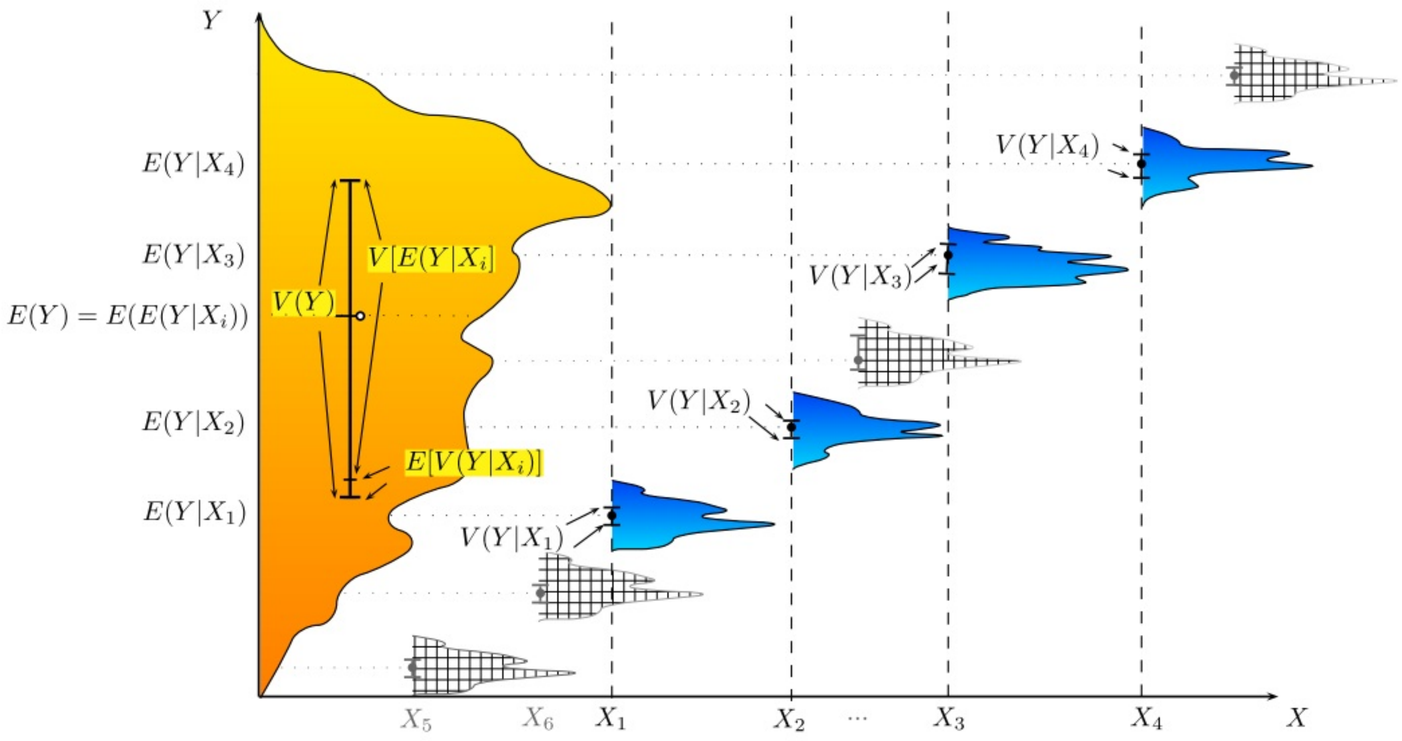


Figure 3: ANOVA : very good fit

Fit by dog breed

https://en.wikipedia.org/wiki/Analysis_of_variance